low intensity, but the meaning he ascribed to these wave shapes was unexpected: *the size of a wave at a given point in space is proportional to the probability that the electron is located at that point in space.* Places where the probability wave is large are locations where the electron is most likely to be found. Places where the probability wave is small are locations where the electron is unlikely to be found. And places where the probability wave is zero are locations where the electron will not be found.

Figure 4.5 gives a "snapshot" of a probability wave with the labels emphasizing Born's probabilistic interpretation. Unlike a photograph of water waves, though, this image could not actually have been made with a camera. No one has ever directly seen a probability wave, and conventional quantum mechanical reasoning says that no one ever will. Instead, we use mathematical equations (developed by Schrödinger, Niels Bohr, Werner Heisenberg, Paul Dirac, and others) to figure out what the probability wave should look like in a given situation. We then test such theoretical calculations by comparing them with experimental results in the following way. After calculating the purported probability wave for the electron in a given experimental setup, we carry out identical versions of the experiment over and over again from scratch, each time recording the measured position of the electron. *In contrast to what Newton would have expected, identical experiments and starting conditions do not necessarily lead to identical measurements.* Instead, our measurements yield a variety of measured locations. Sometimes we find the electron here, sometimes there, and every so often we find it *way* over there. If quantum mechanics is right, the number of times we find the electron at a given point should be proportional to the size (actually, the square of the size), at that point, of the probability wave that we calculated. Eight decades of experiments have shown that the predictions of quantum mechanics are confirmed to spectacular precision.